

Problem Solving (MATH 1000)

Summer 2018

Thursday, 06/07/2018

Quiz 3: Section 10.1, 10.2, 10.3

Name (Print): _____

Score: _____ / 100

1. Write the following percentages as decimal values:

(a) 5 points 0.35%

$$0.35\% = 0.0035$$

(b) 5 points 350%

$$350\% = 3.50$$

2. Write the following decimal values as percentages:

(a) 5 points 0.35

$$0.35 = 35\%$$

(b) 5 points 350

$$350 = 35,000\%$$

3. 15 points Suppose that you deposit \$4,000 in a savings account that pays 2% annual interest, with interest credited to the account at the end of each year. Assuming that no withdrawals are made and that interest is compounded annually, find the balance in the account after 3 years.

The amount of money in the account after 3 years is

$$\begin{aligned} F &= P(1 + r)^t \\ &= \$4,000(1 + 0.02)^3 \\ &= \$4,000(1.02)^3 \\ &= \$4,000(1.061208) \approx \$4,244.83. \end{aligned}$$

4. 15 points For an investment of $P = \$1,262.50$ compounded annually with a 6.75% APR, find the value of the investment after 2 years. Then, find the value of the investment after 3 and a half years.

The value of the investment after 2 years is

$$\begin{aligned} F &= P(1 + r)^t \\ &= \$1,262.50(1 + 0.0675)^2 \\ &= \$1,262.50(1.0675)^2 \approx \$1,438.69. \end{aligned}$$

The value of the investment after 3 and a half years is the same as the value of the investment after 3 years, as the compounding period is yearly. So, the value of the investment after 3 and a half years is

$$\begin{aligned} F &= P(1 + r)^t \\ &= \$1262.50(1 + 0.0675)^3 \\ &= \$1262.50(1.0675)^3 \approx \$1,535.80. \end{aligned}$$

5. 15 points Between the years 1994 and 2004, the average annual inflation rate was 2.7%. Find the salary in the year 2004 that would be equivalent to a \$35,000 salary in 1994.

HINT: If inflation *reduces* the value of your money (relative to some measurement of the average price of things you can buy with a dollar) by 2.7% per year, then your salary needs to *increase* by 2.7% per year in order for it to retain an equivalent value after adjusting for inflation.

Since the annual inflation rate was 2.7%, this means that every year, a dollar became worth 2.7% as much (relative to some measurement of the average price of things you can buy with a dollar). So, for the salary to remain the same with respect to inflation, it would have to increase by 2.7% each year. Thus, the salary in 2004 dollars that would be equivalent to a \$35,000 salary in 1994 is

$$\begin{aligned} F &= P(1 + r)^t \\ &= \$35,000(1 + 0.027)^{10} \\ &= \$35,000(1.027)^{10} \approx \$45,684.88. \end{aligned}$$

6. Congratulations! You just a baby named George! [If you were not expecting this or thought the baby might be named something other than George, also “Surprise!”] You have decided that George needs to go to college, but must have \$40,000 in a savings account before he starts.
- (a) 15 points How much money do you need to put in a savings account that pays 3% compounded quarterly (that is, 4 times per year) so that your George can go to college on his 18th birthday?

We want to determine the principle, P , that we need to deposit so that the future value, F , of the money in the account after 18 years is \$40,000. So, keeping in mind that compounding period is quarterly, the amount of money we need to deposit can be found as follows:

$$\begin{aligned} F &= P(1 + r)^t \\ \$40,000 &= P \left(1 + \frac{0.03}{4} \right)^{(18)(4)} \\ \$40,000 &= P(1.0075)^{72} \\ P &= \frac{\$40,000}{(1.0075)^{72}} \approx \$23,356.95. \end{aligned}$$

- (b) 20 points If you put \$25,000 in a savings account that pays interest compounded monthly, what interest rate is needed so your George can go to college on his 16th birthday?

HINT: $\sqrt[192]{\frac{40,000}{25,000}} \approx 1.00245093421$

We want to determine the interest rate, r (in decimal form), that we need so that a deposit, P , of \$25,000 has a future value, F , after 16 years is \$40,000. So, keeping in mind that compounding period is monthly, the amount of money we need to deposit can be found as follows:

$$\begin{aligned} F &= P(1 + r)^t \\ \$40,000 &= \$25,000 \left(1 + \frac{r}{12}\right)^{(16)(12)} \\ \$40,000 &= \$25,000 \left(1 + \frac{r}{12}\right)^{192} \\ \frac{\$40,000}{\$25,000} &= \left(1 + \frac{r}{12}\right)^{192} \\ \sqrt[192]{\frac{\$40,000}{\$25,000}} &= 1 + \frac{r}{12} \\ \sqrt[192]{\frac{\$40,000}{\$25,000}} - 1 &= \frac{r}{12} \\ r &= 12 \left(\sqrt[192]{\frac{\$40,000}{\$25,000}} - 1 \right) \\ &\approx 0.0294 \end{aligned}$$

So, you need an interest rate of around 2.94% APR to ensure that your \$25,000 investment has a value of \$40,000 after 16 years.